

Directional Derivatives and the Gradient Vector

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Throughout this section, we assume that f is a differentiable function, unless stated otherwise.

Definition If $f(x, y)$ is a function of two variables, then the gradient of is the vector function given by

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle = \frac{\partial f}{\partial x} \vec{i} + \frac{\partial f}{\partial y} \vec{j}$$

Similarly, if $f(x, y, z)$ is a function of three variables, then the gradient of is the vector function given by

$$\nabla f(x, y, z) = \langle f_x, f_y, f_z \rangle = \frac{\partial f}{\partial x} \vec{i} + \frac{\partial f}{\partial y} \vec{j} + \frac{\partial f}{\partial z} \vec{k}$$

Example 1. Find the gradients of the given functions at the indicated points.

- $f(x, y) = 2xy^3 - \frac{x}{y}$ at $P(2, 1)$
- $f(x, y, z) = 4^{x+2z^3} - \ln(x + y)$ at $P(1, 1, 1)$
- $f(x, y, z) = 4x^3 - 6y^2 + 6z + \sin(x^2 + y^3 + z^4)$ at $P(0, \pi, -\pi)$

Recall that, for a function $z = f(x, y)$ we find the partial derivatives at a point (x_0, y_0) By

$$f_x(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h}$$

and

$$f_y(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0, y_0 + h) - f(x_0, y_0)}{h}$$

These partial derivatives represent the rate of change of $f(x, y)$ along the x - and y -axis (i.e. rate of change of $z = f(x, y)$ in the directions of vectors i and j).

We would like to know the rate of change of $z = f(x, y)$ along an arbitrary unit vector u . This brings us to the concept of directional derivatives.

Definition The directional derivative of $z = f(x, y)$ at (x_0, y_0) in the direction of a unit vector $u = \langle a, b \rangle$ is given By

$$D_u f(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + ah, y_0 + bh) - f(x_0, y_0)}{h}$$

provided that the limit exists.

Remark If we take $u = i$ or $u = j$ in the definition above, we would get $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$, respectively.

Using the definition above in computations is not easy. That is why we use the following theorem:

Theorem If f is a differentiable function of x and y , then it has a directional derivative in the direction of any unit vector $u = \langle a, b \rangle$. Furthermore,

$$D_u f(x, y) = f_x(x, y)a + f_y(x, y)b$$

Remark If $z = f(x, y)$ then

$$D_u f(x, y) = \nabla f(x, y) \cdot u$$

Similarly, $w = f(x, y, z)$ then

$$D_u f(x, y, z) = \nabla f(x, y, z) \cdot u$$

Example 2 Find the gradient of f , evaluate the gradient at the point P , and find the rate of change of f at P in the direction of the vector u

- $f(x, y) = y \ln x$, $P(1, -3)$, $u = \langle -\frac{4}{5}, \frac{3}{5} \rangle$
- $f(x, y) = xe^{3xy}$, $P(-1, 3)$, $u = \langle 5, -9 \rangle$
- $f(x, y, z) = \sqrt{x + yz}$, $P(1, 3, 1)$, $u = \langle 2, 3, 6 \rangle$

Maximizing the Directional Derivative

Suppose that f is a function of two or three variables and we consider all possible directional derivatives of f at a given point. These give the rates of change of f in all possible directions.

Question In which direction does f change fastest and what is the maximum rate of change?

Theorem For a function f , the maximum value of the directional derivative $D_u f$ is $|\nabla f|$ and it occurs when u has the same direction as the gradient vector ∇f .

Example 3. In what direction, does f have a maximum rate of change?
What is this maximum rate of change?

- $f(x, y) = ye^{-x} + xe^{-y}$, $P(0, 0)$
- $f(x, y, z) = \ln(xy^2z^3)$, $P(1, -2, -3)$
- $f(x, y) = \cos(xy)$, $P(1, 0)$

Example 4.

On a certain mountain, the elevation z above a point (x, y) in an xy -plane at sea level is $z = 2000 - 0.02x^2 - 0.04y^2$, where x , y , and z are in meters. The positive x -axis points east, and the positive y -axis north. A climber is at the point $(-20, 5, 1991)$.

- If the climber uses a compass reading to walk due west, will she begin to ascend or descend?
- If the climber uses a compass reading to walk northeast, will she ascend or descend? At what rate?

Example 5.

A mosquito is flying in a greenhouse where the humidity at any point is given by the equation

$$H(x, y, z) = \frac{1}{2} \sin(xy) + \frac{1}{z+1}$$

- Which direction should the mosquito fly from its current position of $P(\pi, 2, 1)$ to decrease the humidity of the air around the fastest?
- How does the humidity change if the mosquito instead flies in the direction of the point $Q(\pi + 3, 2, 5)$? How does the humidity change in this direction?

Now consider a surface given by the equation $F(x, y, z) = k$ where $k \in \mathbb{R}$. Suppose that $P(x_0, y_0, z_0)$ is a point on this surface and $C : r(t) = \langle x(t), y(t), z(t) \rangle$ is a curve on this surface. Clearly we have

$$F(x(t), y(t), z(t)) = k$$

Differentiate both sides wrt t (using the chain rule) to get

$$\frac{\partial F}{\partial x} \frac{dx}{dt} + \frac{\partial F}{\partial y} \frac{dy}{dt} + \frac{\partial F}{\partial z} \frac{dz}{dt} = 0$$

Since $\nabla F = \langle F_x, F_y, F_z \rangle$ and $r'(t) = \langle x'(t), y'(t), z'(t) \rangle$ we have

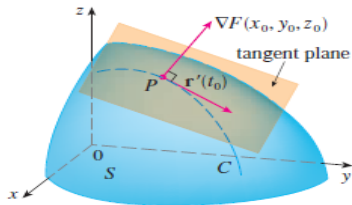
$$\nabla F \cdot r'(t) = 0$$

If $t = t_0$ corresponds to the point P , then we have

$$\nabla F(x_0, y_0, z_0) \cdot r'(t_0) = 0$$

Thus, $\nabla F(x_0, y_0, z_0) \perp$ the tangent vector to any curve the surface passing through $P(x_0, y_0, z_0)$.

Definition The **tangent plane** to the surface $F(x, y, z) = k$ at $P(x_0, y_0, z_0)$ is the plane that passes through P and has normal vector $\nabla F(x_0, y_0, z_0)$.



Definition The **normal line** to a surface at a point P is the line passing through P and perpendicular to the tangent plane.

Proposition Consider a surface given by a function $F(x, y, z) = k$. The equation of the tangent plane to S at $P(x_0, y_0, z_0)$ is

$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0$$

The equation of the normal line to S at $P(x_0, y_0, z_0)$ is

$$\frac{x - x_0}{F_x(x_0, y_0, z_0)} = \frac{y - y_0}{F_y(x_0, y_0, z_0)} = \frac{z - z_0}{F_z(x_0, y_0, z_0)}$$

Proof.

As a particular case, consider

$z = f(x, y) \implies F(x, y, z) = f(x, y) - z = 0$. Therefore

$\nabla F = \langle f_x, f_y, -1 \rangle$. Hence, the equation of the tangent plane at $P(x_0, y_0)$ becomes

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - x_0) - (z - z_0) = 0$$

Example 6. Find the equations of the tangent planes and normal lines to

- $x = y^2 + z^2 - 2$ at $P(-1, 1, 0)$
- $x - z = 4 \tan^{-1}(yz)$ at $P(1 + \pi, 1, 1)$
- $yz = 4 \ln(x + z)$ at $P(0, 0, 1)$

Example 7. Show that the ellipsoid $3x^2 + 2y^2 + z^2 = 9$ and the sphere $x^2 - 8x + y^2 - 6y + z^2 - 8z = -24$ are tangent to each other at the point $(1, 1, 2)$.

Example 8. Show that every plane that is tangent to the surface $z^2 = x^2 + y^2$ passes through the origin.

Example 9. Find parametric equations for the tangent line to the curve of intersection of the paraboloid $z = x^2 + y^2$ and the ellipsoid $4x^2 + y^2 + z^2 = 9$ at the point $(-1, 1, 2)$.

Example 10. Show that the function $f(x, y) = \sqrt[3]{xy}$ is continuous and the partial derivatives f_x and f_y exist at the origin but the directional derivatives in any other directions do not exist.

Remark.

$f(x, y)$ is differentiable at $(a, b) \implies D_u f(a, b)$ exists in every direction (a, b) .

Proof.

Suppose that $f(x, y)$ is differentiable at (a, b) . $\implies$

$$f(a + h, b + k) = f(a, b) + f_x(a, b)h + f_y(a, b)k + o\left(\sqrt{h^2 + k^2}\right).$$

Let

$$\mathbf{u} = \langle u_1, u_2 \rangle$$

be a unit vector $\implies$

$$D_{\mathbf{u}} f(a, b) = \lim_{t \rightarrow 0} \frac{f(a + tu_1, b + tu_2) - f(a, b)}{t}.$$

Since f is differentiable,

$$D_{\mathbf{u}} f(a, b) = f_x(a, b)u_1 + f_y(a, b)u_2$$

or, equivalently,

$$D_{\mathbf{u}} f(a, b) = \nabla f(a, b) \cdot \mathbf{u}.$$

Therefore,

Differentiability $\implies$ directional derivatives exist in every direction.

The existence of directional derivatives in every direction does not guarantee differentiability.

Example Consider

$$f(x, y) = \begin{cases} \frac{x^3}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Let $u = \langle \cos \theta, \sin \theta \rangle$ be any unit direction. $\implies$

$$f(t \cos \theta, t \sin \theta) = \frac{(t \cos \theta)^3}{(t \cos \theta)^2 + (t \sin \theta)^2} = t \cos^3 \theta.$$

Therefore,

$$\begin{aligned} D_u f(0, 0) &= \lim_{t \rightarrow 0} \frac{f(t \cos \theta, t \sin \theta) - f(0, 0)}{t} \\ &= \lim_{t \rightarrow 0} \frac{t \cos^3 \theta}{t} \\ &= \cos^3 \theta. \end{aligned}$$

Thus the directional derivative exists for every direction:

$$D_u f(0, 0) = \cos^3 \theta$$

for every θ .

On the other hand,

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{h - 0}{h} = 1$$

Similarly,

$$f_y(0, 0) = \lim_{h \rightarrow 0} \frac{f(0, h) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0.$$

$$\Rightarrow \nabla f(0, 0) = \langle 1, 0 \rangle.$$

The linearization at $(0, 0)$ is

$$L(x, y) = f(0, 0) + f_x(0, 0)x + f_y(0, 0)y = x.$$

For differentiability we must have

$$\lim_{(x,y) \rightarrow (0,0)} \frac{f(x, y) - f(0, 0) - L(x, y)}{\sqrt{x^2 + y^2}} = \lim_{(x,y) \rightarrow (0,0)} \frac{f(x, y) - x}{\sqrt{x^2 + y^2}} = 0.$$

However, along the path $y = x$, we have

$$\lim_{(x,y) \rightarrow (0,0)} \frac{f(x, y) - x}{\sqrt{x^2 + y^2}} = \lim_{x \rightarrow 0} \frac{\frac{x}{2} - x}{\sqrt{2x^2}} = \lim_{x \rightarrow 0} \frac{-x/2}{\sqrt{2}|x|} = -\frac{1}{2\sqrt{2}} \lim_{x \rightarrow 0} \frac{x}{|x|} \quad DNE.$$

Therefore, f is not differentiable at $(0, 0)$.

Example II. In which direction is the derivative of $f(x, y) = xy + y^2$ at the point $P(3, 2)$ equal to zero?

Solution. Note that

$$\nabla f = \langle y, x + 2y \rangle \implies \nabla f(3, 2) = \langle 2, 7 \rangle.$$

Thus, vectors orthogonal to $\nabla f(3, 2)$ are $\vec{u}_1 = \langle 7, -2 \rangle$ and $\vec{u}_2 = \langle -7, 2 \rangle$.

Example 12. Suppose that $\vec{u} = \langle a, b \rangle$ is a unit vector f and has continuous second partial derivatives. Prove that

$$D_{\vec{u}}^2 f = a^2 f_{xx} + 2abf_{xy} + b^2 f_{yy}$$

Then, use this formula to find the **second directional derivative** of the function $f(x, y) = e^x y^3$ in the direction of the vector $\vec{u} = \langle 5, 12 \rangle$.

Remark. $\nabla f(x, y)$ vs $\nabla F(x, y, z)$

Let $z = f(x, y)$ be a function, define $F(x, y, z) := z - f(x, y)$.

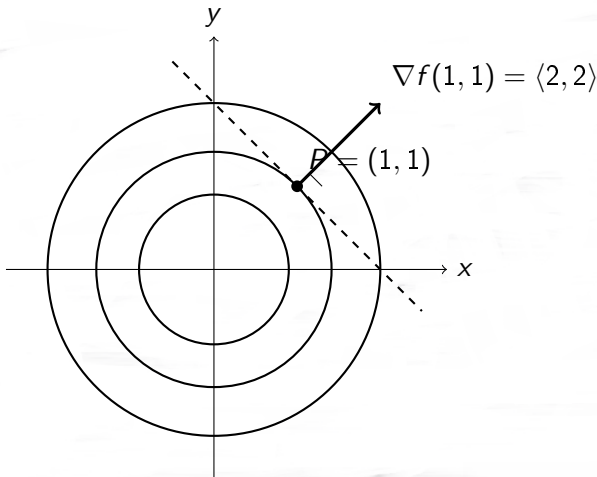
- $\nabla f(x, y)$ is perpendicular to every level curve to $f(x, y) = c$ in the xy -plane.
- $\nabla F(x, y, z)$ is perpendicular to the surface defined by $z = f(x, y)$ in the xyz -space.

Example. Let $f(x, y) = x^2 + y^2$ and $P = (1, 1)$. Then the level curves are the circles

$$x^2 + y^2 = c$$

and

$$\nabla(f)(x, y) = \langle 2x, 2y \rangle \implies \nabla(f)(1, 1) = \langle 2, 2 \rangle$$



Let $F(x, y, z) := z - f(x, y) = z - x^2 - y^2$. then $\nabla(F)$ at the same point looks like this:

